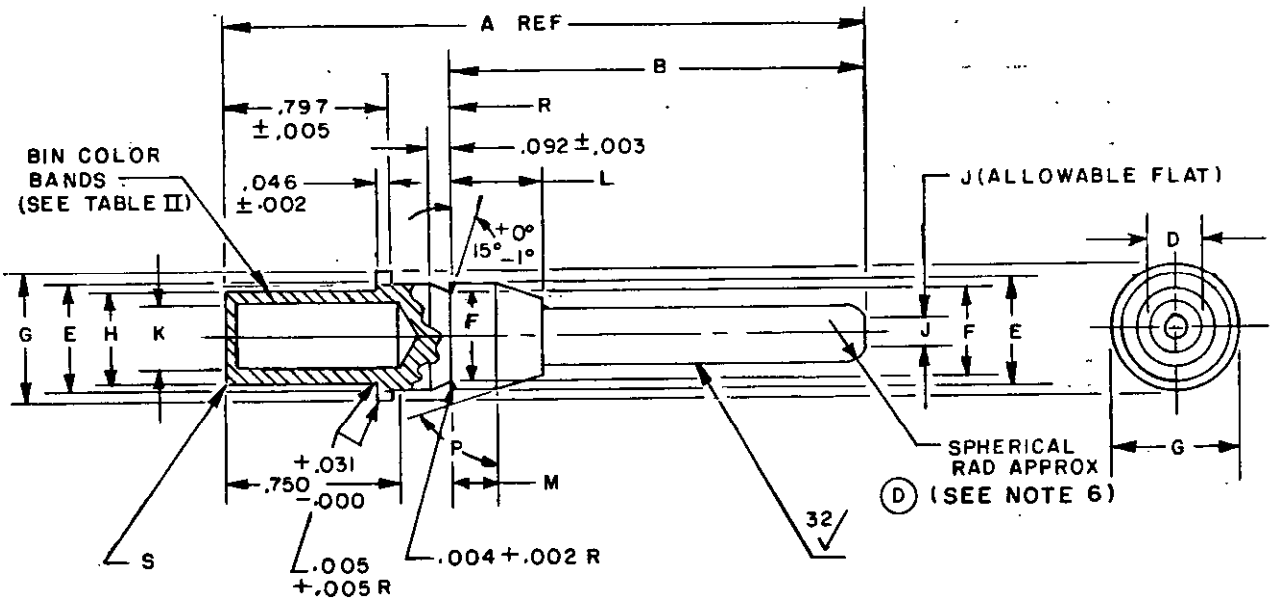
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, PIN, CRIMP REMOVABLE, (FOR MIL-C-22992 CLASS L CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.



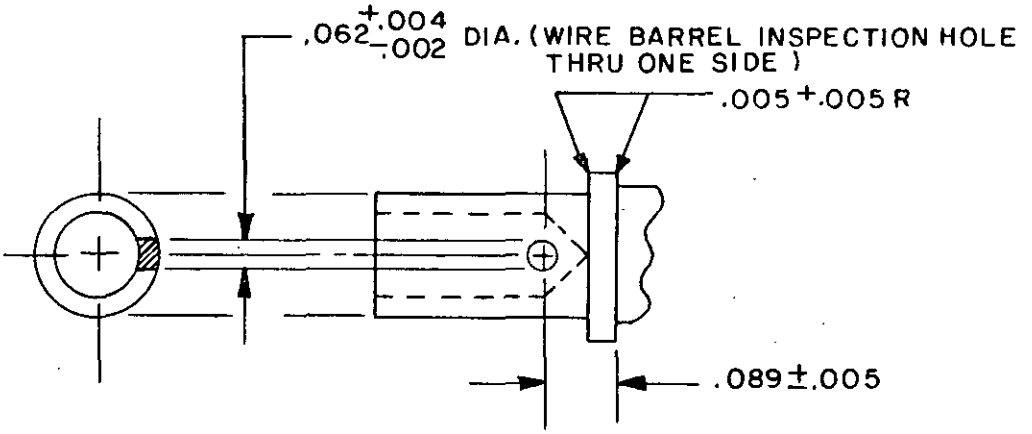
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. At indicated area, hardness to be as listed in table.
5. Unless otherwise specified, tolerances are ±.010 and ±2°.
Ⓓ 6. Spherical end dimension shall conform to MIL-C-39029.
7. To properly crimp a wire of a smaller gauge than the contact wire barrel, the appropriate MS 3348 bushing must be used.
8. Manufacturer's trademark location optional.

FIGURE 1. Dimensions and configuration.

Ⓓ denotes changes

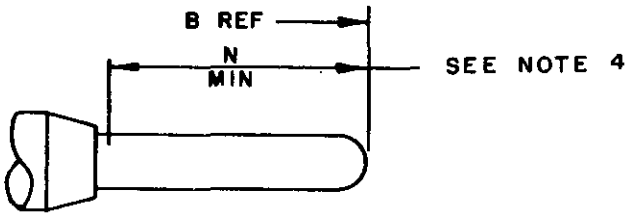
FSC 5935

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VIEW A

WIRE BARREL (BORE BARREL CRIMP DEPTH)



BIN code	N MIN	ROCKWELL HARDNESS MIN
317	1.406	F80
318	1.531	
319	1.375	
320	1.406	
321	1.531	
322	1.375	
323	1.719	OPTIONAL
324	1.844	
325	1.719	
326	1.844	
237	1.719	
328	1.844	

INCHES	MM
1.375	34.93
1.406	35.71
1.531	38.89
1.719	43.66
1.844	46.84

FIGURE 1. Dimensions and configuration - Continued.

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	0.05	.078	1.98	.232	5.89	.334	8.48	.506	12.85	1.738	44.15
.003	0.08	.089	2.26	.234	5.94	.342	8.69	.516	13.11	1.746	44.35
.004	0.10	.092	2.34	.250	6.35	.344	8.74	.609	15.47	1.856	47.14
.005	0.13	.100	2.54	.251	6.38	.357	9.07	.641	16.28	2.097	53.26
.006	0.15	.113	2.87	.281	7.14	.374	9.50	.654	16.61	2.215	56.26
.015	0.38	.178	4.52	.286	7.26	.375	9.52	.688	17.48	2.786	70.76
.016	0.41	.179	4.55	.297	7.54	.406	10.31	.748	19.00	2.856	72.54
.020	0.51	.188	4.78	.308	7.82	.417	10.59	.750	19.05	2.904	73.76
.031	0.79	.203	5.16	.312	7.92	.490	12.45	.781	19.84	3.207	81.46
.046	1.17	.225	5.72	.313	7.95	.500	12.70	.797	20.24	3.325	84.46
.062	1.57										

TABLE I. Dimensions

BIN code	A REF	B ±.005	D DIA ±.001	E DIA +.002 -.001	F DIA		G DIA		H DIA		J DIA ±.010	K DIA		L ±.005	M ±.005	P	R ±.002	S MIN 60° SLOPE
317	2.786	1.738	.178	.312	.286	±.002	.342	±.002	.312	±.002	.078	.234	±.002	.312	.188	75°	.251	.015
318	2.904	1.856	.178	.312	.286		.342		.312		.078	.234		.312	.188	75°	.251	.015
319	2.856	1.746	.178	.312	.286		.342		.312		.078	.234		.344	.297	60°	.313	.015
320	2.786	1.738	.225	.334	.308		.417		.374		1/	.281		.312	.188	75°	.251	.015
321	2.904	1.856	.225	.334	.308		.417		.374		1/	.281		.312	.188	75°	.251	.015
322	2.856	1.746	.225	.334	.308		.417		.374		1/	.281		.344	.297	60°	.313	.015
323	3.207	2.097	.357	.516	.490	+.002 -.005	.609	±.003	.506	+.004 -.003	1/	.406	+.004 -.003	.344	.297	60°	.313	.020
324	3.325	2.215	.357	.516	.490		.609		.506		1/	.406		.344	.297	60°	.313	.020
325	3.207	2.097	.406	.654	.629		.688		.609		1/	.500		.344	.297	60°	.313	.020
326	3.325	2.215	.406	.654	.629		.688		.609		1/	.500		.344	.297	60°	.313	.020
327	3.207	2.097	.500	.748	.723		.781		.750		1/	.641		.344	.297	60°	.313	.020
328	3.325	2.215	.500	.748	.723		.781		.750		1/	.641		.344	.297	60°	.313	.020

1/ Maximum allowable. Pin tip flat in accordance with MIL-C-39029.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Dimensions, design characteristics, and configuration: See figure 1 and tables I and II.

Wire barrel: Wire barrel ranges and size shall be a specified in table IV.

Tools: See table III.

Vibration: In accordance with method 2005 of MIL-STD-1344, test condition III. The following detail and exceptions shall apply.

a. Normal locking means only, no safety wire shall be used.

b. The cable or wire bundle shall be clamped to nonvibrating points at least 8 inches from the rear of the connectors. The clamping length may be selected or changed to avoid resonance of the cable or wire.

c. The system ground contacts of class L grounding connectors shall not be wired into the series monitoring circuit.

Shock: MIL-C-39029/48 contacts shall be tested in accordance with the drop test (class L) and high impact shock test as specified in MIL-C-22992, using a MIL-C-22992 qualified connector that has been approved for listing on the applicable qualified products list.

Mating contact: MIL-C-39029/49.

Material: Copper alloy SAE CA145; SAE CA147. Hard temper; CABRA 147 or equivalent.

Finish: Silver plate .0002 inch thick minimum in accordance with QQ-S-365. Use underplate of .00005 inch minimum nickel in accordance with QQ-N-290.

Concentricity: .004 TIR(RFS), machined surfaces 63 $\overline{V}$ unless otherwise specified.

Contacts may be soldered or crimped.

Ⓓ Qualification: Shall include the crimping of these contacts utilizing MS3448 bushings to accommodate smaller wire sizes.

QPL evaluating activity: Defense Electronics Supply Center (DESC-E), Dayton, Ohio 45444.

International interest: NEPR 57, class L only.

Ⓓ Military part number: See table V.

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TABLE II. Design characteristics.

Bin code	Color bands			Mating end size	Wire barrel size	Type	Class
	1st	2nd	3rd				
317	Orange	Brown	Violet	6	6		
318	Orange	Brown	Gray	6N 1/	6		
319	Orange	Brown	White	6G 1/	6		
320	Orange	Red	Black	4	4		
321	Orange	Red	Brown	4N 1/	4		
322	Orange	Red	Red	4G 1/	4	A	A
323	Orange	Red	Orange	1/0	1		
324	Orange	Red	Yellow	1/0N 1/	1		
325	Orange	Red	Green	2/0	2/0		
326	Orange	Red	Blue	2/0N 1/	2/0		
327	Orange	Red	Violet	4/0	4/0		
328	Orange	Red	Gray	4/0N 1/	4/0		

1/ "N" designation indicates a neutral pin. "G" designation indicates a grounding pin.

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Ⓓ TABLE III. Tools.

Bin code	Basic crimping tool	Locator	Die assembly	Removal tool
317 through 319	M22520/23-01	M22520/23-10	M22520/23-03	M81969/27-06
320 through 322	M22520/23-01	M22520/23-12	M22520/23-04	M81969/27-05
323 and 2/ 324	M22520/23-01	M22520/23-14	M22520/23-05	M81969/27-03
325 and 326	M22520/23-01	M22520/23-15	M22520/23-06	M81969/27-02
327 and 2/ 328	M22520/23-01	M22520/23-16	M22520/23-07	M81969/27-01

1/ Installing tools are not necessary with those contacts.

2/ When crimping contacts using a MS3348-1-6L or MS3348-4/0-2/0L bushings use the 400-B-1 tool only.

Ⓓ TABLE IV. Allowable wire barrel - wire size combinations.

Bin code	Wire barrel size	Allowable wire size	Required bushing
317, 318, 319	6	6	----------
		8	MS3348-6-8L
		9	MS3348-6-9L
		10	MS3348-6-10L
320, 321, 322	4	4	----------
		5	MS3348-4-5L
		6	MS3348-4-6L
		8	MS3348-4-8L
323 324	1	1	----------
		2	MS3348-1-2L
		6	MS3348-1-6L
325 326	2/0	2/0	----------
327 328	4/0	4/0	----------
		2/0	MS3348-4/0-2/0L

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Ⓓ TABLE V. Military part number.

Bin code	Military part number	Supersedes
317	M39029/48-317	MS90559-11
318	M39029/48-318	MS90559-12
319	M39029/48-319	MS90559-14
320	M39029/48-320	MS90559-8
321	M39029/48-321	MS90559-9
322	M39029/48-322	MS90559-13
323	M39029/48-323	MS90559-5
324	M39029/48-324	MS90559-6
325	M39029/48-325	MS90559-3
326	M39029/48-326	MS90559-4
327	M39029/48-327	MS90559-1
328	M39029/48-328	MS90559-2

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, MC
Air Force - 11, 17, 99
DLA - ES

User activities:
Navy - OS, SH
Army - AT

Agent:
DLA - ES

Preparing activity:
Navy - EC

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